



Masters Thesis Poster

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Bacterial Active Turbulence, and Polymeric Influence on it - An Experimental Study

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Background & Motivation

Experimental studies of bacterial turbulence

- Weisink et al. 2012 *PNAS* → quasi 2D, *B. subtilis*
- Wei et al. 2024 *Adv. Sci.* → 2D to 3D, *E. coli*

Computational study of bacterial turbulence

- Mukherjee et al. 2023 *Nat. Phys.* → 2D (Scaling of $E(k) \sim k^{-1.5}$)

Comparison with inertial turbulence

- No universal scaling exponent for energy spectra, unlike inertial turbulence: $E(k) \sim k^{-5/3}$ (Kolmogorov 1941 *Cr. Acad. Sci. USSR*)
- No calculations of scaling exponents in intermittency analyses, unlike inertial turbulence: $\kappa \sim R^{-0.1}$ (Herweijer & van de Water 1995 *PRL*, Anselmet et al. 1984 *Jr. Fluid Mech.*)

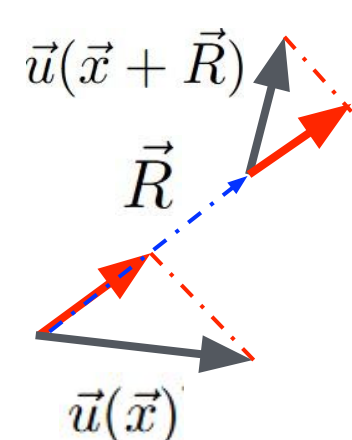
Objectives of this work

(1) Revisit experiments on active turbulence of *E. coli*: obtain detailed statistics of flow velocities

(2) How do the statistics change on homogenous addition of polymer (Hydroxyethyl Cellulose) in small amount?

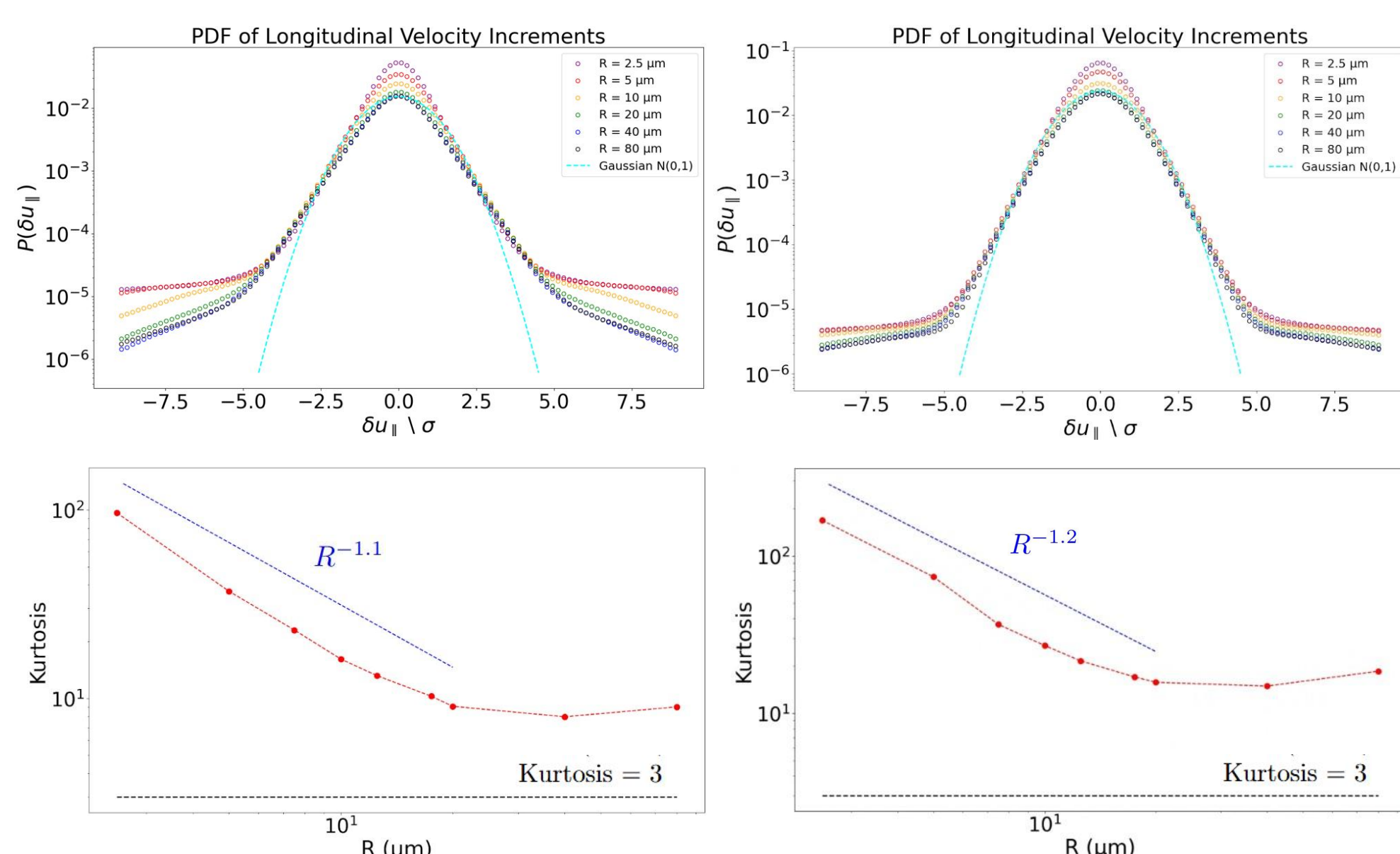
Results For Polymer-Free Bacterial Turbulence

Spatial Velocity Fluctuations: Intermittency



$$\delta u_{\parallel} = [\vec{u}(\vec{x} + \vec{R}) - \vec{u}(\vec{x})] \cdot \frac{\vec{R}}{|\vec{R}|}$$

$$\text{Kurtosis } \kappa = \frac{\langle \delta u_{\parallel}^4 \rangle}{\langle \delta u_{\parallel}^2 \rangle^2}$$

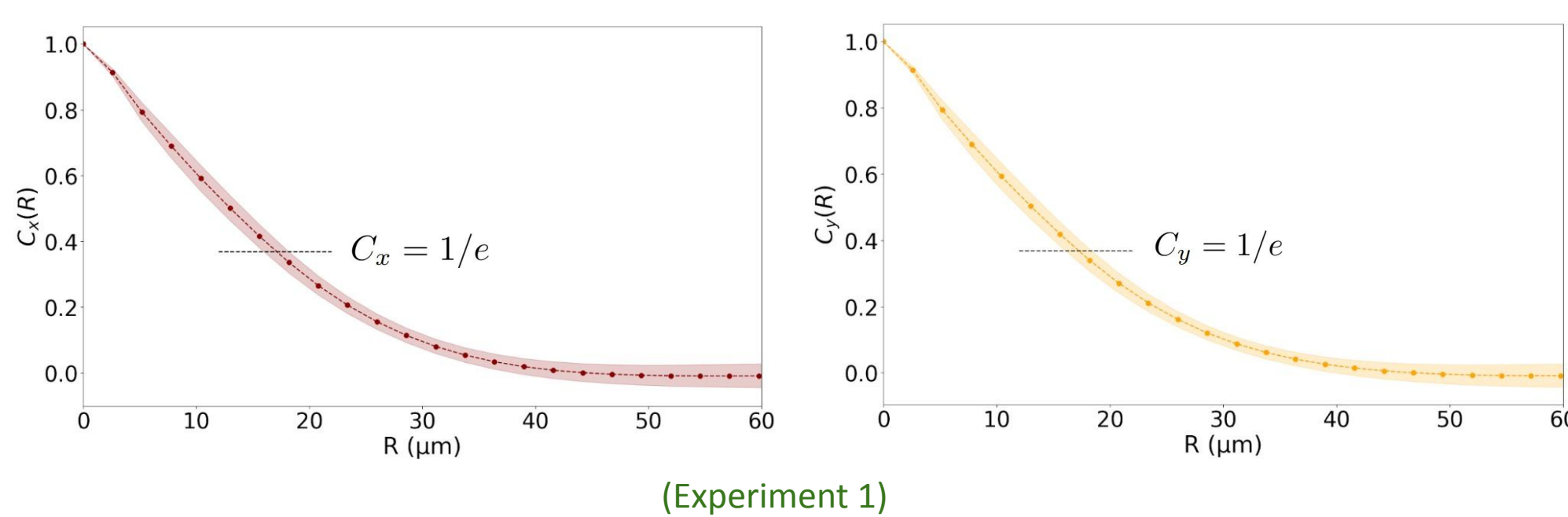


(Experiment 1)

(Experiment 2)

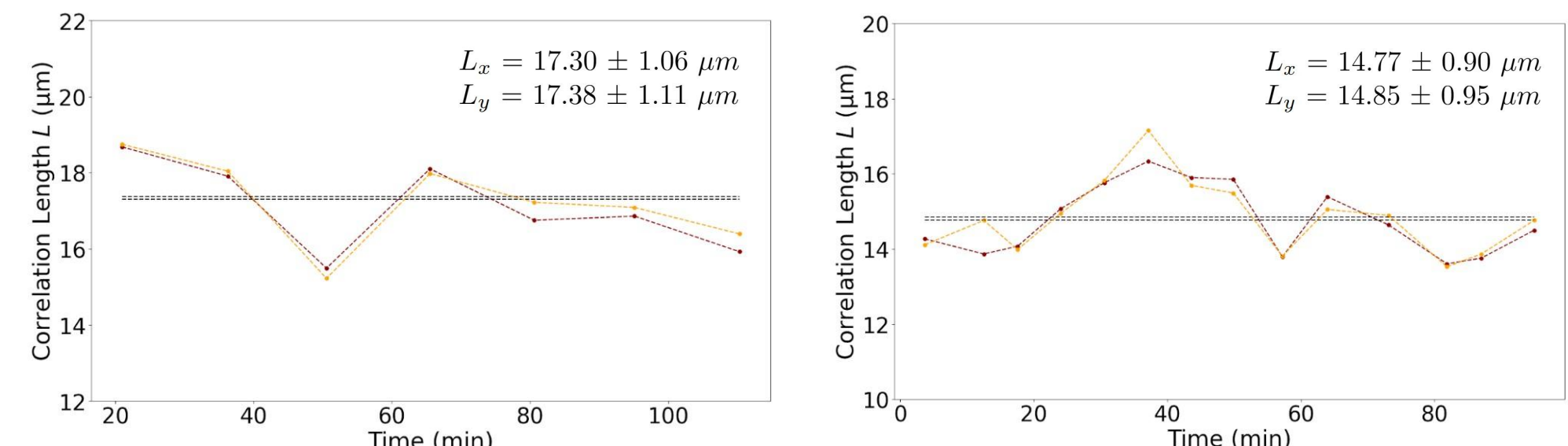
Spatial Velocity Correlations

$$C_x(R) = \frac{\langle \vec{u}(x, y) \cdot \vec{u}(x + R, y) \rangle}{\langle \vec{u}(x, y) \cdot \vec{u}(x, y) \rangle} \quad C_y(R) = \frac{\langle \vec{u}(x, y) \cdot \vec{u}(x, y + R) \rangle}{\langle \vec{u}(x, y) \cdot \vec{u}(x, y) \rangle} \quad R, \text{ when } C = 1/e \rightarrow L$$



(Experiment 1)

Proof of structural steady state

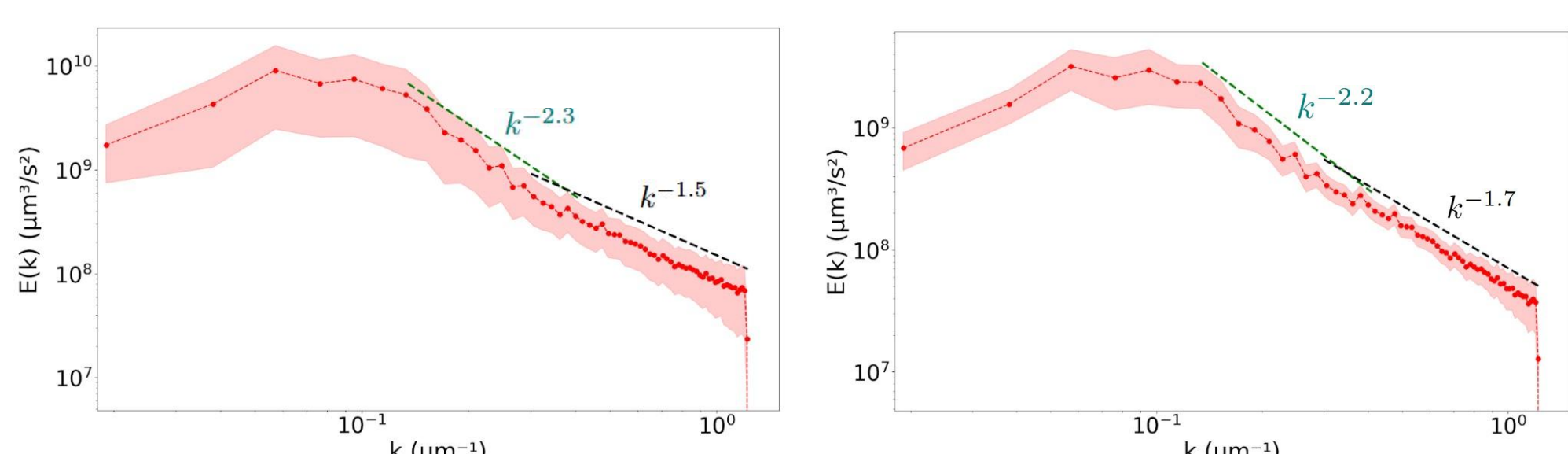


(Experiment 1)

(Experiment 2)

Energy Spectra

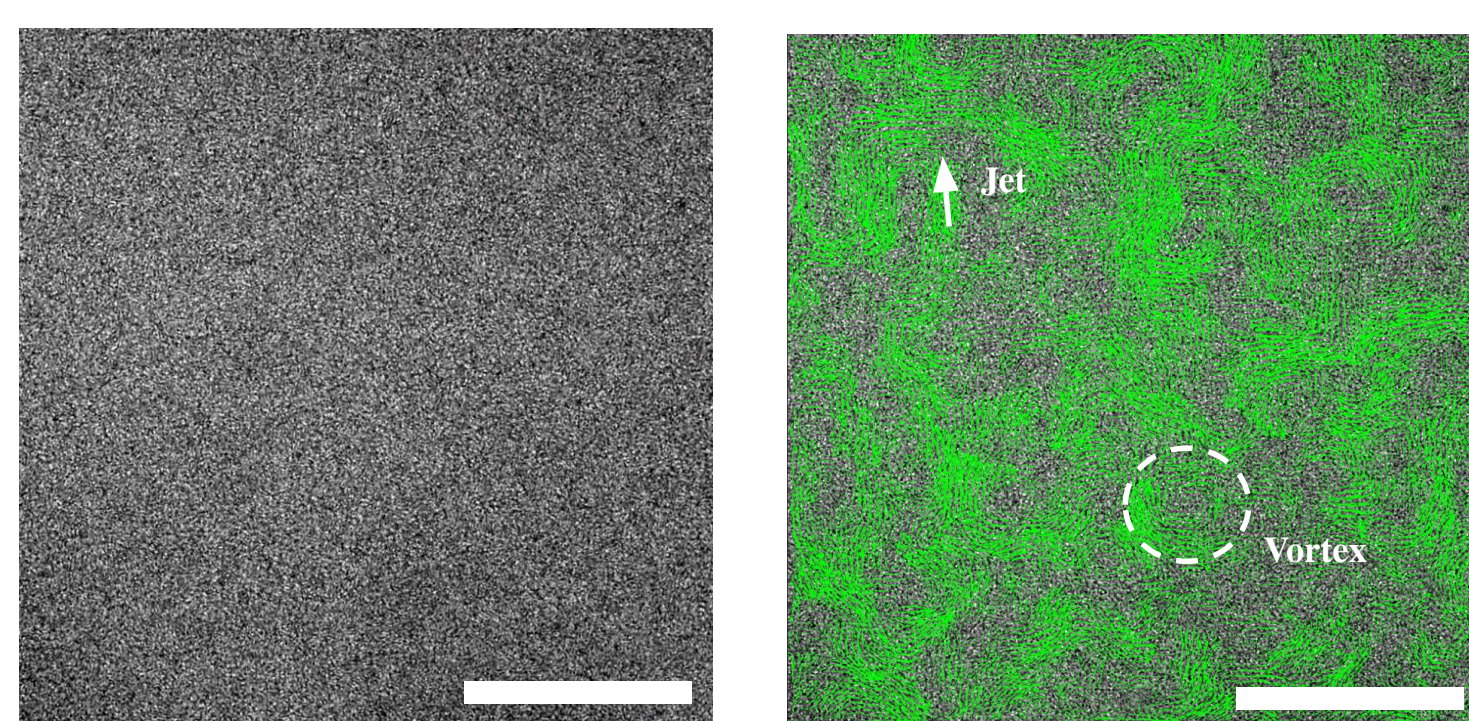
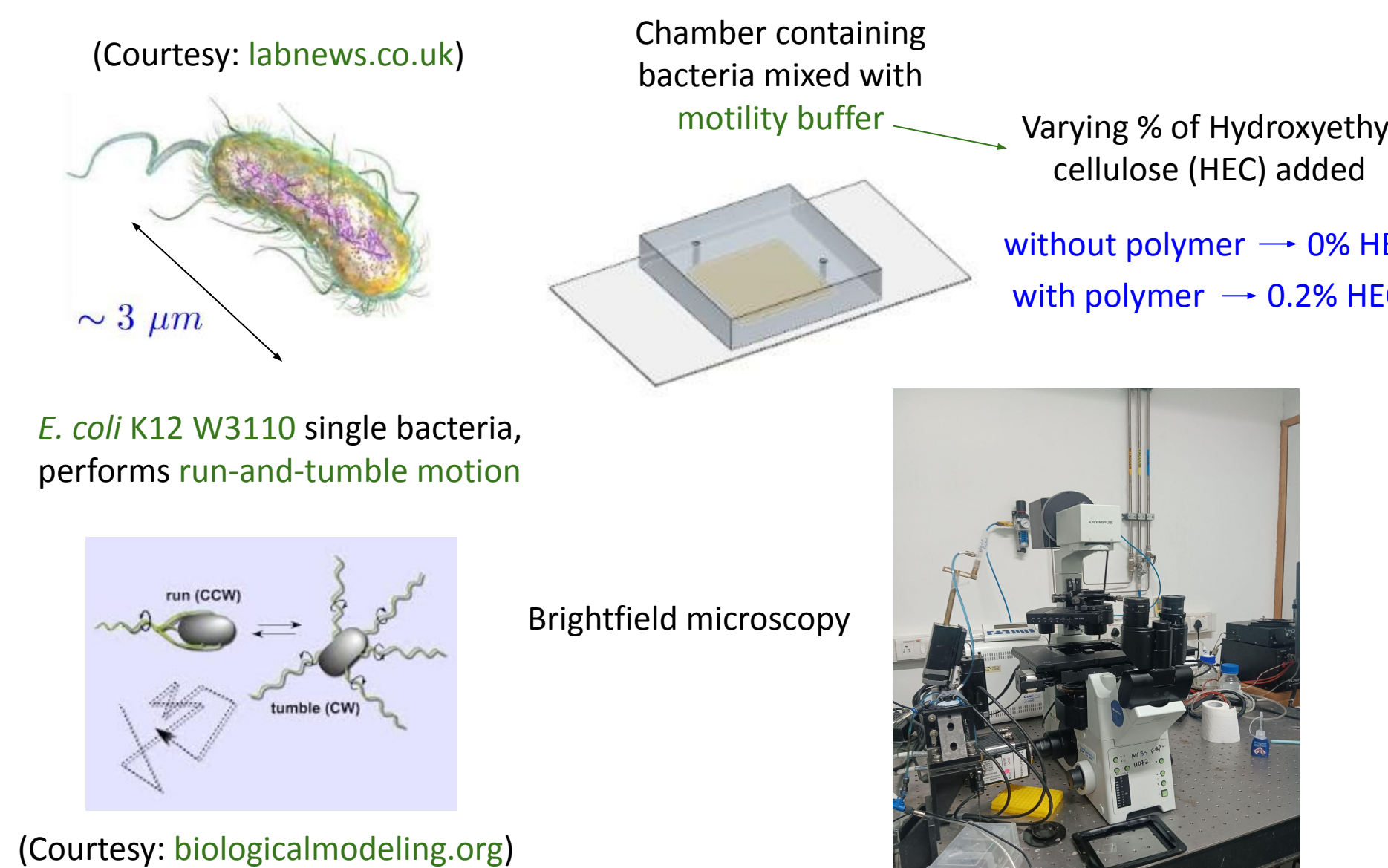
Velocity field in real space $(u(x, y), v(x, y))$ $\xrightarrow{\text{F.T.}}$ Velocity field in Fourier space $(\vec{u}(k_x, k_y), \vec{v}(k_x, k_y))$ $\rightarrow E(k) \propto (|\vec{u}|^2 + |\vec{v}|^2)$
 $k = \sqrt{k_x^2 + k_y^2}$



(Experiment 1)

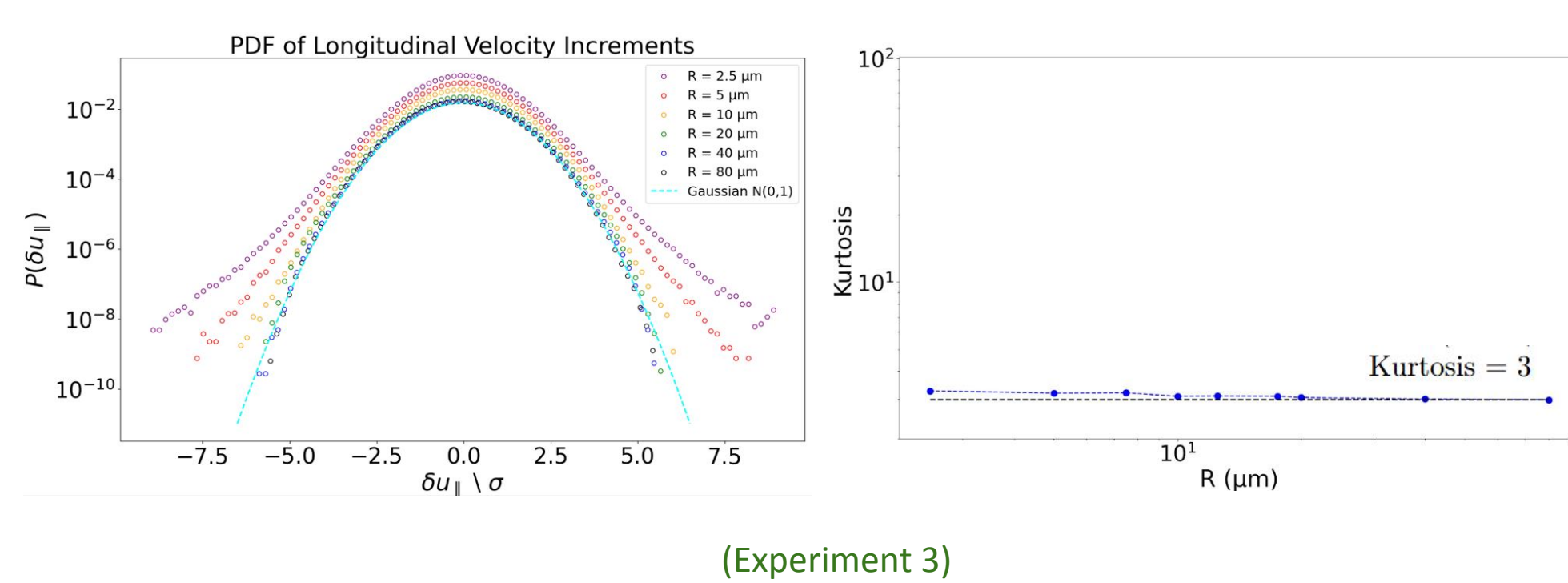
(Experiment 2)

Bacterial Description, Experimental Setup & Velocity Flows

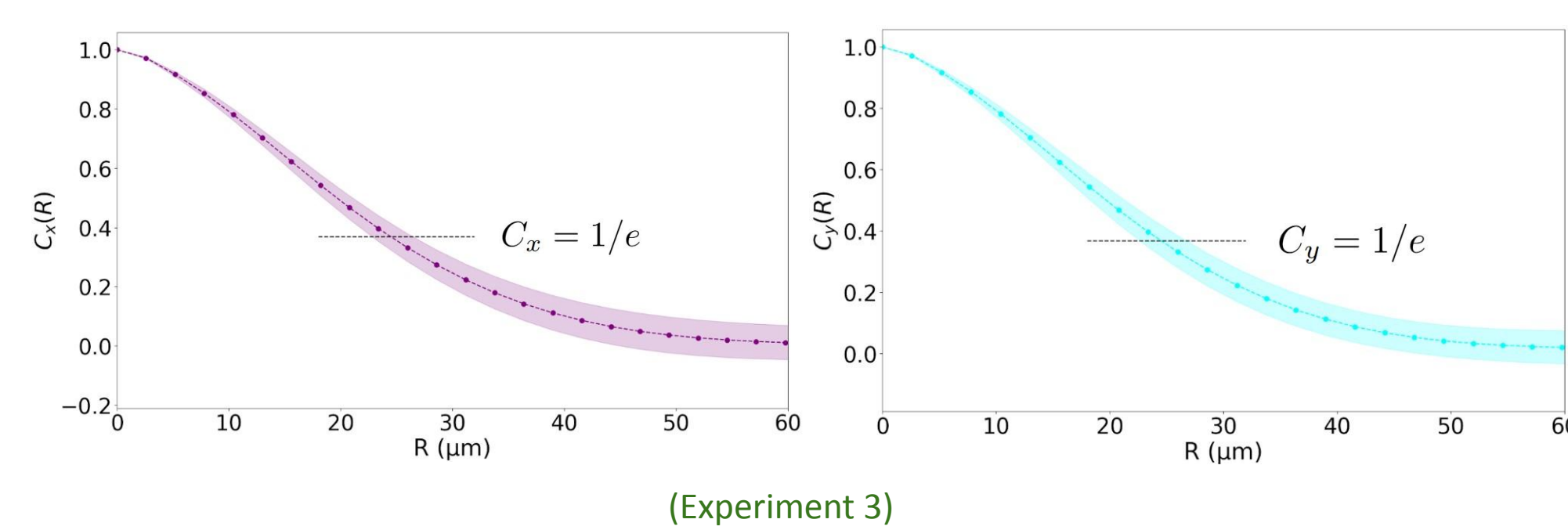


Results For Polymer-Added Bacterial Turbulence

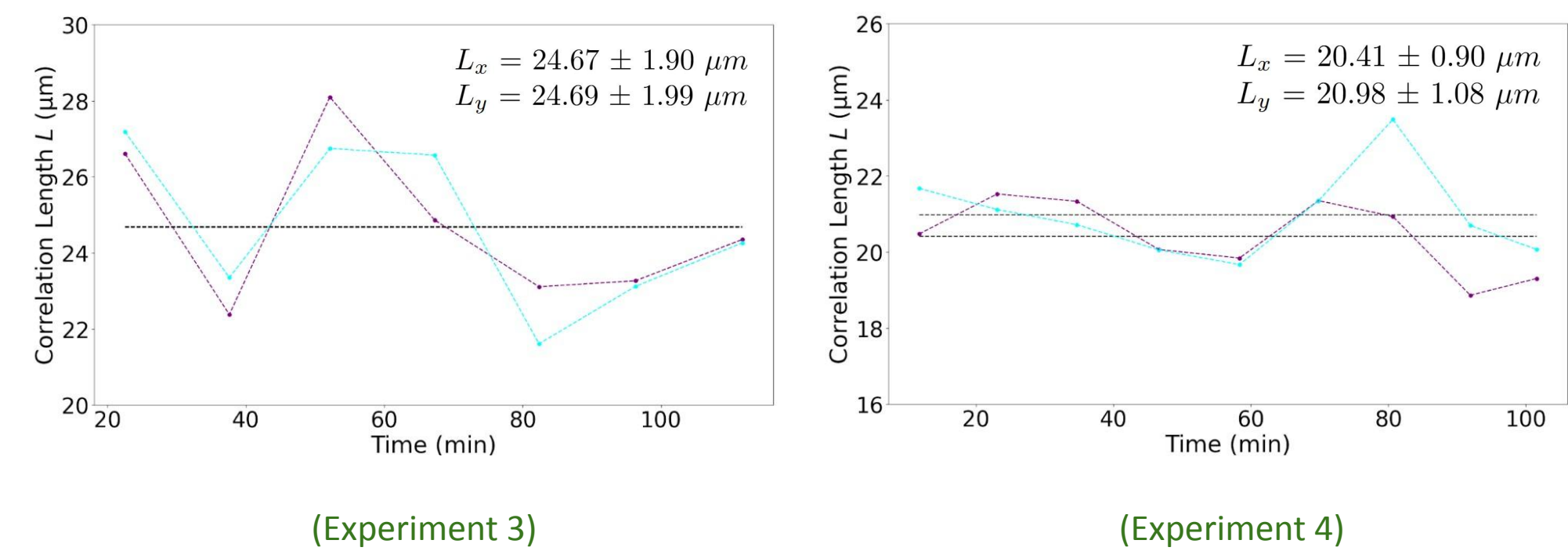
Spatial Velocity Fluctuations: Intermittency



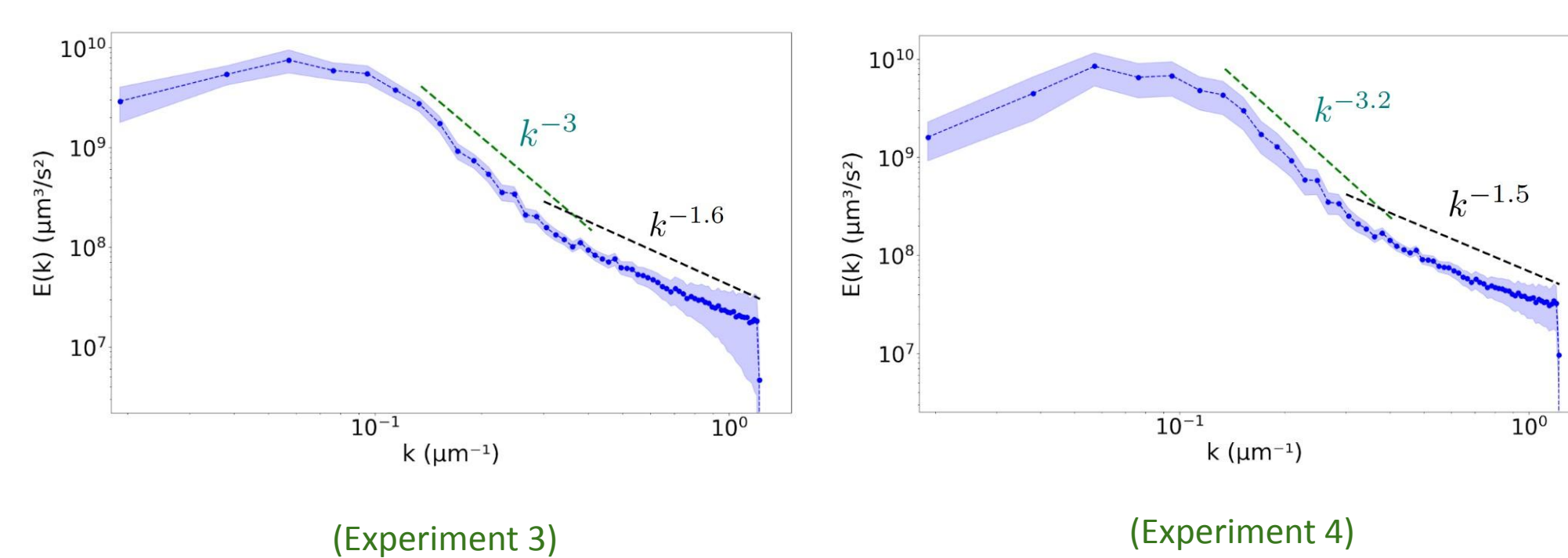
Spatial Velocity Correlations



Proof of structural steady state



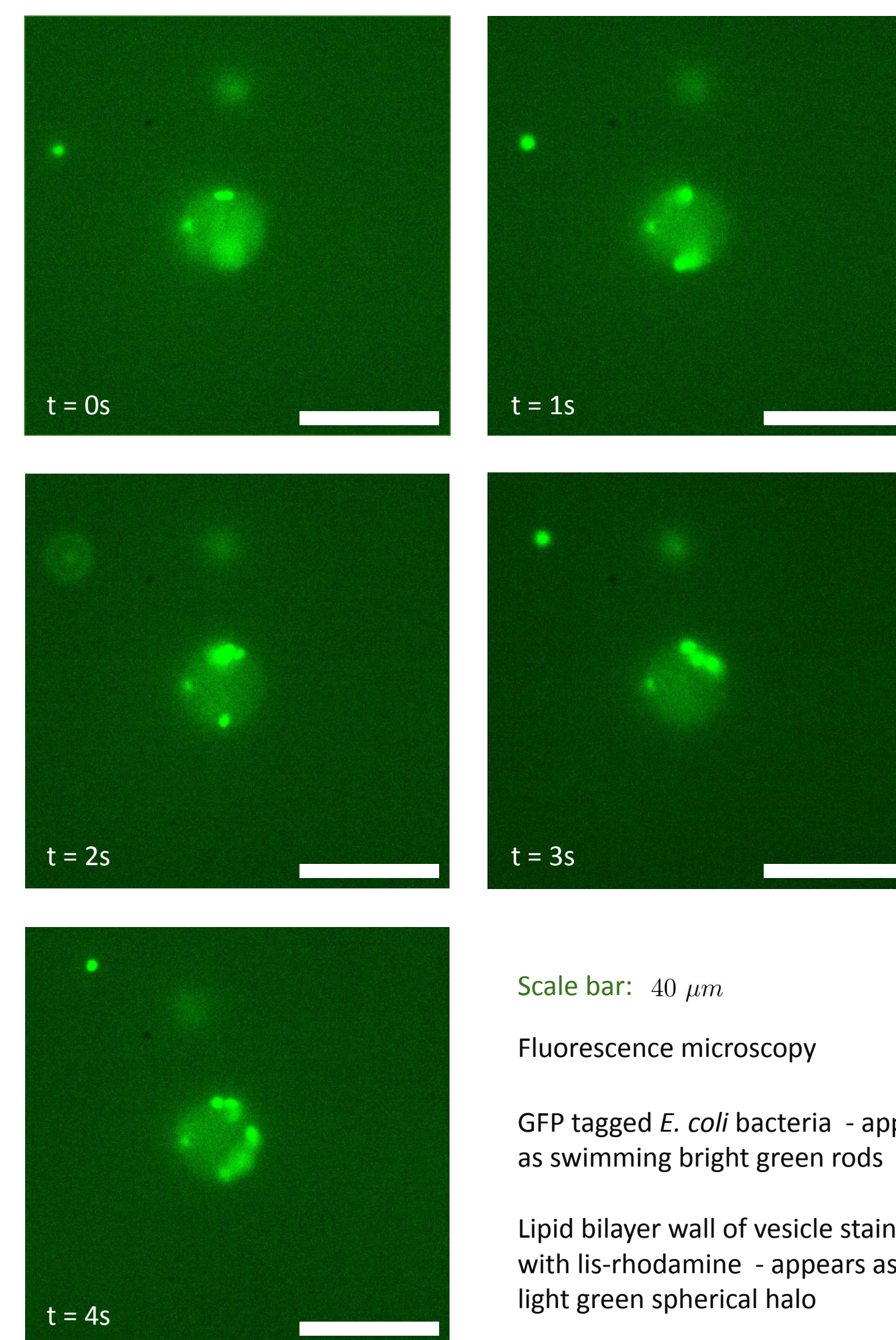
Energy Spectra



Summary

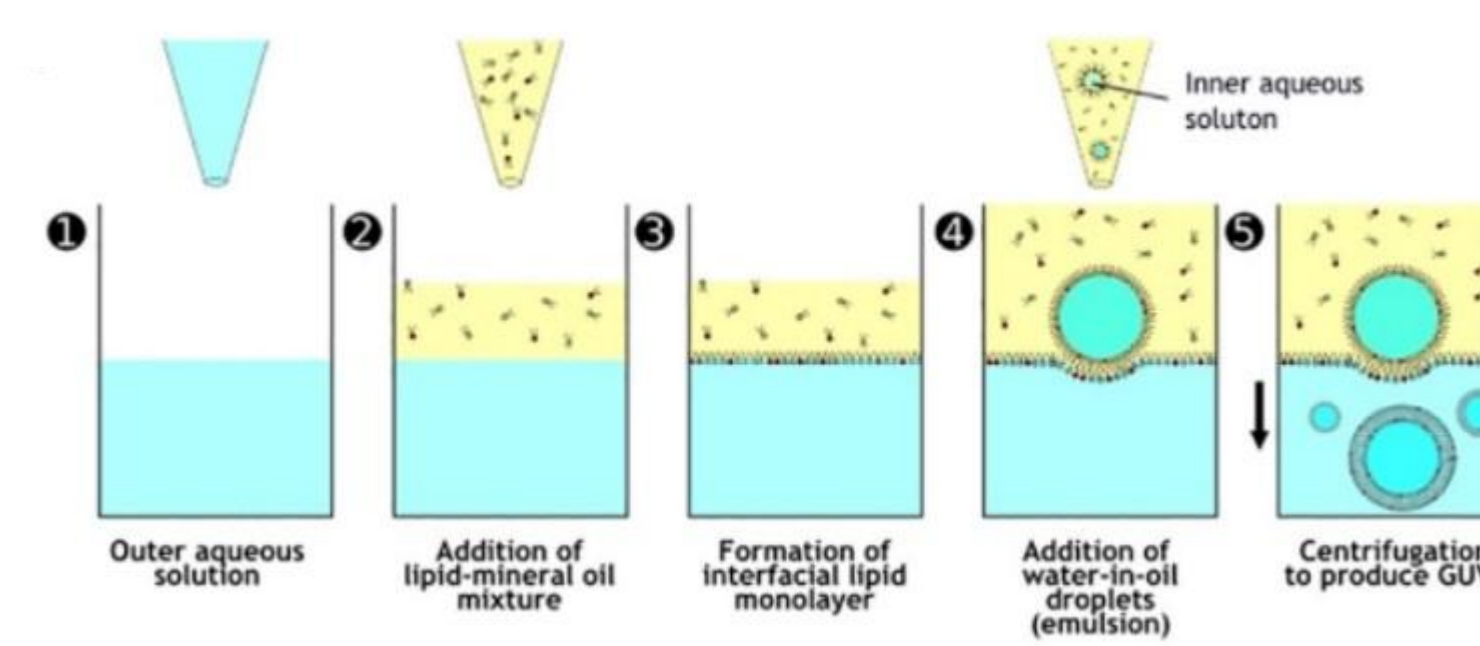
- E. coli* Active Turbulence
 - Spatial velocity fluctuations are intermittent: Kurtosis goes as $R^{-1.15 \pm 0.05}$ (much higher exponent than in inertial turbulence)
 - $E(k)$ has power law scalings: $k^{-2.25 \pm 0.05}$ & $k^{-1.6 \pm 0.1}$
- Polymer addition in Bacterial Turbulence
 - Reduce spatial velocity fluctuations: Intermittency vanishes (Kurtosis ~ 3)
 - Make flows more correlated (correlation lengths are systematically higher)
 - Makes energy transfer across intermediate length-scales more inefficient (power law scaling in intermediate k regime: $k^{-3.1 \pm 0.1}$)

Side Project: Trapping Bacteria in Lipid Vesicles



Experimental Preparation

(Courtesy: Moga et al. 2019 *ChemBioChem*)



Outer solution: LB containing bacteria + 400 mM of sucrose + 0.025% SDS

Inner solution: 840 mM of glucose

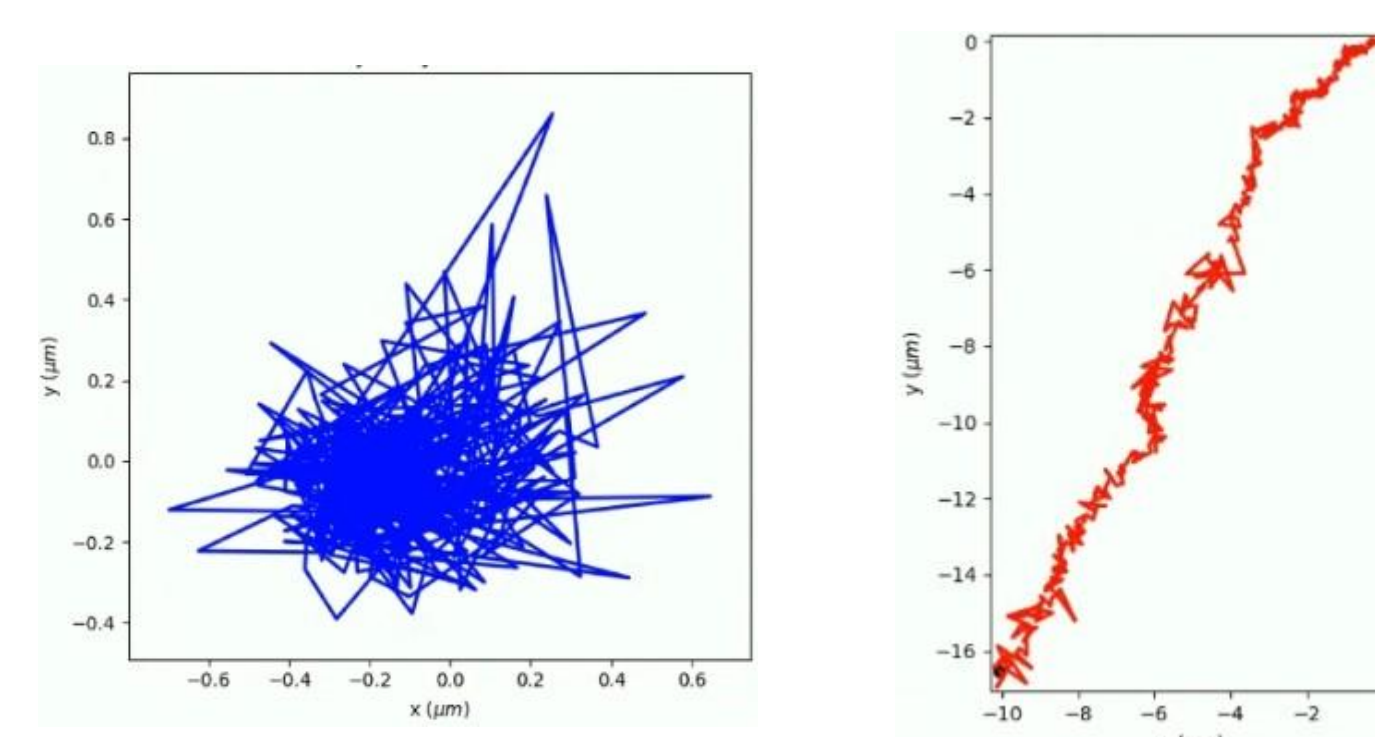
Oil: Mineral oil

Lipid: DOPC (2 mg/ml)

Centrifugation speed & time: 1500 g, 10 mins

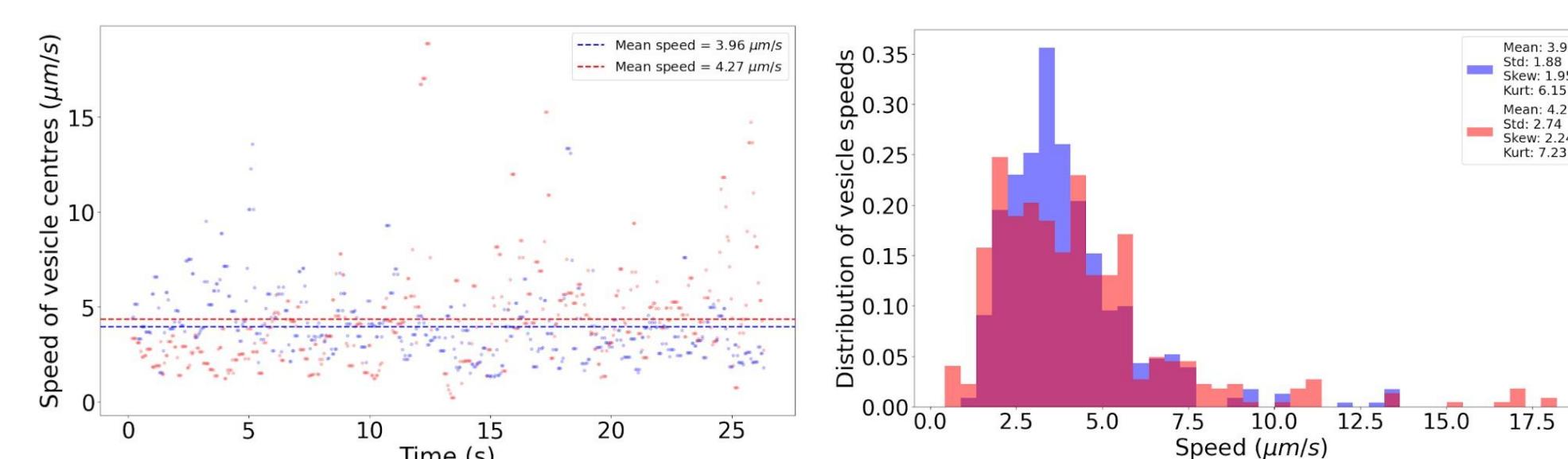
Preliminary Analysis of these Vesicles

Tracking vesicle centers reveal different kinds of trajectories



Calculated mean speeds are much higher than those previously reported in literature

(Le Nagard et al. 2022 *PNAS*)



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